
BAYESIAN COMPARISON OF BEC, NFW, AND gNFW HALO MODELS USING GALAXY ROTATION CURVES

ABSTRACT

We present a Bayesian comparison of three dark-matter halo models—Navarro–Frenk–White (NFW), generalized NFW (gNFW, with free inner slope α), and Bose–Einstein condensate (BEC) halos including a rotating variant—fitted to galaxy rotation curves from the SPARC database to address the core–cusp problem. Unlike earlier studies that relied on heuristic or genetic-algorithm searches of parameter space, we sample the full posterior distributions with Markov Chain Monte Carlo, marginalizing over baryonic nuisance parameters and using both the Bayesian Information Criterion (BIC) and a posterior-based harmonic-mean evidence estimator to assess model support. Across the sample most galaxies are statistically indistinguishable between models, but allowing a free inner slope yields a modest, systematic evidence gain for gNFW relative to fixed-slope NFW. BEC models show a much broader spread in performance: a subset of galaxies strongly favour BEC, but many exhibit substantial evidence loss compared to gNFW, with the rotating BEC giving the best mean performance yet large galaxy-to-galaxy variability. These results highlight that flexibility in the inner slope is crucial for rotation-curve fits, and demonstrate that full Bayesian evidence estimation provides a more robust, population-level assessment of halo models than previous genetic-algorithm or point-estimate approaches.

Contents

1	Introduction	2
2	Methods	3
2.1	Theoretical Framework and Mass Distribution	3
2.2	Observational Data	4
2.3	Mass Model Fitting	4
2.4	Generalized NFW (gNFW) Dark Matter Halo Model	4
2.5	BEC Dark Matter Halo Model	5
2.6	Non-Rotating BEC	5
2.7	Rotating BEC	5
2.8	Bayesian Inference	6
2.9	Bayesian Model Comparison	6
2.10	BIC Approximation	6
2.11	Harmonic–Mean Estimator	6
2.12	Model Parameters and Priors	7
3	Results	8

3.1 Posterior Distribution Analysis	8
3.2 Model Comparison Metrics	11
4 Discussion	12
5 Conclusions	12

1 Introduction

Dark matter is an invisible form of matter that neither emits nor absorbs electromagnetic radiation. Although it cannot be observed directly, it makes up 85 percent of the total matter in the universe and plays a critical role in shaping cosmic structure. Its presence is inferred solely through gravitational influence on visible matter, light, and the universe's large-scale structure. Admittedly, there is sufficient evidence pointing to this unseen mass component which dominates the dynamics of galaxies and clusters [1].

The very first evidence of dark matter came when Fritz Zwicky studied the Coma galaxy cluster and he noticed that the galaxies were moving much faster than what would be expected based on just the visible mass. After applying the virial theorem, he concluded that the cluster must contain far more mass than what was observable and he coined the term 'dark matter'. The discrepancy has since been confirmed and further extended by other observations.

One of the most compelling lines of evidence comes from the study of galactic rotation curves as Newtonian physics predicts that the orbital velocities of stars would decrease with increased distance from the galactic center just as planets in the solar system which is further described in the methods section [2]. However, observations show that these velocities remain flat even far from the galactic bulge. This unexpected result implies that galaxies are surrounded by massive, extended halos of dark matter, which provide the extra gravitational pull needed to keep stars in orbit.

Gravitational lensing also reveals the presence of dark matter. The bending of light depends on total mass, not just visible mass, allowing astronomers to "weigh" galaxies and clusters. In many cases, the observed lensing cannot be explained by luminous matter alone. A particularly dramatic example is the Bullet Cluster, a collision of two galaxy clusters in which the luminous gas (visible in X-rays) lags behind the main gravitational mass, suggesting that the majority of matter is non-interacting and invisible. On even larger scales, the cosmic microwave background (CMB) offers a snapshot of the early universe.

Detailed measurements of the CMB's temperature fluctuations by satellites such as WMAP and Planck have allowed cosmologists to extract precise information about the universe's composition. The amount of normal (baryonic) matter is actually insufficient to explain the gravitational effects observed; instead, a significant contribution from cold, non-baryonic dark matter is required to match the observed power spectrum. Finally, simulations of structure formation in the universe rely on dark matter to explain how galaxies and clusters emerged from tiny initial fluctuations.

The Navarro–Frenk–White (NFW) profile, derived from cold dark matter simulations [3], has become a standard description of galactic dark matter halos. However, the NFW form is fixed to an inner logarithmic slope of $\alpha = 1$, which may not adequately capture the diversity of observed galaxy rotation curves. To address this, the generalized NFW (gNFW) profile introduces a free slope parameter α , modifying the inner density law. This additional degree of freedom allows the gNFW profile to interpolate between cored halos ($\alpha \approx 0$) and steeper cusps ($\alpha > 1$), making it a more flexible baseline model.

The core-cusp problem arises from the discrepancy between simulations and observations of dark matter density profiles in galaxies. N-body simulations based on CDM predict a steep density increase towards the center of galaxies (a "cusp"), described by the Navarro-Frenk-White (NFW) profile. However, observations of galaxy rotation curves often suggest a constant-density core in the inner regions of galaxies. This contradiction indicates the need to explore alternative dark matter models that can naturally produce cored density profiles.

One such alternative is Bose-Einstein Condensate (BEC) dark matter. A BEC is a state of matter that occurs at extremely low temperatures, where a large fraction of bosons occupy the lowest quantum state, resulting in macroscopic quantum phenomena [4]. In the context of dark matter, it is hypothesized that ultra-light bosons could form a galactic-scale condensate, leading to a smooth, core-like density distribution. This model has the potential to resolve the core-cusp discrepancy while maintaining consistency with large-scale structure formation.

In this study, we analyze a rotating BEC dark matter halo model [5] and fit it to observational data. Unlike previous models that considered only static halos, rotation introduces a more realistic dynamical behavior and alters the predicted rotation curves.

For our analysis, we made use of the SPitzer Photometry and Accurate Rotation Curves (SPARC) dataset [6], which provides high-quality rotation curves for a large sample of nearby galaxies spanning a wide range of morphological types, luminosities, and surface brightnesses. Each galaxy in SPARC includes both observed kinematic data and carefully derived baryonic contributions from stellar and gas components, enabling detailed mass modeling. In this work, we fit these rotation curves with a set of competing dark matter halo profiles, extracting posterior distributions for their parameters and comparing their Bayesian evidences across the sample. This framework allows us to statistically evaluate which models are most consistent with the diversity of observed galaxy dynamics.

In this work, we treat gNFW as the reference model when comparing alternatives such as Bose–Einstein condensate (BEC) halos, since it subsumes NFW as the special case $\alpha = 1$ and provides a direct test of whether observed galaxies favor shallower or steeper inner slopes. This approach also links directly to the long-standing “core–cusp problem” in astrophysics, where low-mass galaxies often appear to prefer cored dark matter distributions in tension with the canonical NFW prediction.

Aside from the astrophysical motivations, this work will also be addressing a crucial methodological gap as previous studies examining BEC halos have not incorporated rotation in the condensate model or lack a Bayesian framework for statistical comparison. By using Bayesian inference, we may obtain full posterior distributions for each parameter as well as the uncertainties or standard deviations [7]. In addition, we access the Bayesian evidence which provides us with a principled measure of relative model support.

While the BIC provides a quick approximation to the Bayesian evidence using only the maximum likelihood and the number of parameters, it ignores the full shape of the posterior. In this work, we go beyond $\ln Z_{\text{BIC}}$ by also evaluating the harmonic–mean estimator, which uses the complete set of posterior samples from MCMC [8]. This allows the evidence calculation to reflect the integrated support across parameter space rather than being tied only to the best–fit point, offering a more faithful comparison between halo models despite the known noisiness of the harmonic mean approach.

Rotation curves serve as an especially powerful probe in this context. Unlike lensing, which predominantly constrains cluster-scale structures, or the cosmic microwave background, which reflects cosmological scales, rotation curves directly trace the distribution of mass within galaxies. This makes them uniquely sensitive to the inner halo structure where the long-standing core–cusp tension between observations and Λ CDM predictions arises. By comparing rotating BEC halos with NFW and gNFW models, our analysis connects directly to this debate, testing whether quantum condensate models can naturally reproduce the cored profiles suggested by low-mass galaxies, or whether classical cuspy models remain statistically preferred.

2 Methods

2.1 Theoretical Framework and Mass Distribution

The study of galaxy rotation curves relies on the assumption that stars and gas in the galaxies orbit the center in a circular path. Under Newtonian gravity, the balance between the gravitational and centripetal forces gives:

$$\frac{v(r)^2}{r} = \frac{GM(r)}{r^2} \quad (1)$$

Solving for the orbital velocity $v(r)$ at a radius r :

$$v(r) = \sqrt{\frac{GM(r)}{r}} \quad (2)$$

Here, G is the universal gravitational constant and $M(r)$ is the mass that is enclosed within the radius r . If the mass distribution followed just the visible matter, $M(r)$ would then approach a constant value at a large value of r , and the velocity would decline consequently as $1/\sqrt{r}$. Observed rotation curves, however, remain flat or can even rise at large radii. This implies that $M(r)$ continues to grow even as little luminous matter can be seen beyond the radius. This discrepancy suggests the presence of dark matter.

To compute $M(r)$, we can integrate the mass density profile $\rho(r)$ over a spherical volume:

$$M(r) = \int_0^r 4\pi r'^2 \rho(r') dr' \quad (3)$$

To calculate the enclosed mass the integral of the mass density is evaluated numerically using the trapezoidal rule [9]. This method involves dividing the radial range into small intervals and approximating the area under the curve by averaging the values of density at the endpoints of each interval. It offers computational simplicity making it effective for this task.

2.2 Observational Data

The rotation curves for galaxies are constructed using Doppler spectroscopy. In the inner galaxy optical emission lines like H α can be used [10]. In the outer regions they are traced using the 21-cm line from neutral hydrogen (HI). These measurements provide velocities of gas and stars, which are then corrected for the inclination angle of the galaxy relative to earth in order to derive the circular velocities about the galactic center. By plotting velocity as a function of radius, the rotation curve of the galaxy is obtained.

We employ data from the SPitzer Photometry and Accurate Rotation Curves (SPARC) database [11] which contains rotation curves for 175 galaxies. The sample spans a wide range of luminosities, mass and morphological types.

2.3 Mass Model Fitting

To interpret the rotation curve, we model the galaxy's mass as a combination of visible (baryonic) components and a dark matter halo. The total rotation velocity is given by:

$$v_{\text{total}}^2(r) = v_{\text{disk}}^2(r) + v_{\text{bulge}}^2(r) + v_{\text{halo}}^2(r) \quad (4)$$

The luminous components are derived from surface brightness profiles, scaled by an assumed mass-to-light ratio. The dark matter halo is typically modeled using a pseudo-isothermal profile:

$$\rho_{\text{halo}}(r) = \frac{\rho_0}{1 + \frac{r^2}{r_c^2}} \quad (5)$$

So the velocity contributed by the halo can be described as:

$$v_{\text{halo}}(r) = \frac{S}{4\pi G \rho_0 r_c^2} \left[1 - \frac{r_c}{r} \tan^{-1} \frac{r}{r_c} \right] \quad (6)$$

Here, ρ_0 is the central density and r_c is the core radius of the halo. These parameters can then be adjusted to best fit the observed rotation curve using least-squares minimization.

2.4 Generalized NFW (gNFW) Dark Matter Halo Model

A widely used model for dark matter halos is the Navarro-Frenk-White (NFW) profile, derived from cold dark matter (CDM) simulations. It has the form:

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{\frac{r}{r_s} \left(1 + \frac{r}{r_s} \right)^2}, \quad (7)$$

where ρ_s is a characteristic density and r_s is the scale radius. While successful in many contexts, the NFW profile has a central cusp ($\rho \propto r^{-1}$) that often disagrees with observed galaxy cores.

To allow for more flexibility, we use the *generalized* NFW (gNFW) profile, which introduces an inner slope parameter α :

$$\rho_{\text{gNFW}}(r) = \frac{\rho_s}{\frac{r}{r_s}^\alpha \left(1 + \frac{r}{r_s} \right)^{3-\alpha}}. \quad (8)$$

Here, ρ_s denotes the characteristic density of the halo and r_s the scale radius. The parameter α controls the inner slope of the halo: $\alpha = 1$ recovers the standard NFW profile, while values $\alpha < 1$ correspond to a cored profile and $\alpha > 1$ produce a steeper cusp. In practice, α is constrained by both simulations and observations to lie roughly in the range $0 \leq \alpha \leq 1.5$, allowing the gNFW form to interpolate between shallow cores and steep cusps. This flexibility makes it well suited for addressing the “core-cusp problem,” a long-standing tension between CDM simulations, which favor cuspy halos, and dwarf galaxy rotation curves, which often suggest cores.

The gNFW model is particularly useful because it interpolates between cored and cuspy halos, enabling a more general comparison against observed rotation curves. In our Bayesian framework, α is treated as a free parameter, and its posterior distribution across galaxies serves as a diagnostic of halo structure. One of the key outputs of our analysis is the distribution of best-fitting α values, summarized in a histogram across the sample. This directly reveals whether galaxies statistically favor NFW-like cusps ($\alpha \sim 1$) or core-like structures ($\alpha < 1$), and provides an intuitive way to visualize the degree of tension between simulations and observations.

2.5 BEC Dark Matter Halo Model

In this work, we model the dark matter halo as a self-gravitating Bose-Einstein Condensate (BEC). A BEC is a state of matter in which a large fraction of bosons occupy the ground quantum state, allowing the system to be described by a single macroscopic wavefunction. In the astrophysical context, such a condensate forms when ultra-light bosonic dark matter particles (e.g., axions) thermalize and condense under gravity.

2.6 Non-Rotating BEC

Assuming a spherically symmetric condensate under the Thomas-Fermi approximation, the non-rotating density profile is given by:

$$\rho(r) = \rho_c \frac{\sin(\pi r/R)}{\pi r/R}, \quad (9)$$

where ρ_c is the central density and R is the radius at which the density drops to zero (i.e., the edge of the halo). This form predicts a density core with a finite central value, which makes the non-rotating BEC halo particularly relevant for addressing the observed core-cusp problem in dwarf galaxies. The corresponding velocity profile is obtained from the enclosed mass $M(r)$ of the condensate and is set by the two free parameters ρ_c and R .

The sinusoidal functional form reflects the underlying quantum nature of the condensate, as the density profile arises from the ground-state solution of the Gross-Pitaevskii-Poisson equations under self-gravity. This naturally produces a central core in contrast to the CDM simulations.

This non-rotating case provides a useful baseline against which the impact of rotation can be evaluated. Any deviation introduced by the rotational degree of freedom highlights the dynamical richness of the condensate model, motivating its comparison to cuspy profiles such as NFW and the more flexible gNFW.

2.7 Rotating BEC

To incorporate rotation, we assume the condensate rotates as a rigid body with angular velocity ω . This introduces a centrifugal term that modifies the effective gravitational potential and alters the density distribution. In the first-order approximation, the tangential (circular) velocity profile for the rotating BEC halo is given by:

$$v_{\text{tg}}^2(r) = \frac{4G\rho_c R^2}{\pi} (1 - \Omega^2) \frac{\sin(\pi r/R)}{\pi r/R} - \cos \frac{\pi r}{R} + \frac{\Omega^2}{3} \frac{\pi r^2}{R}. \quad (10)$$

where: In this expression, $v_{\text{tg}}(r)$ denotes the circular velocity at radius r , and G is the gravitational constant. The parameter ρ_c represents the central density of the condensate, while R is the halo radius, defined as the point where the density drops to zero. Finally, Ω is a dimensionless parameter characterizing rotation, given by $\Omega = \omega/\omega_0$, where $\omega_0 = \sqrt{G\rho_c}$ is the characteristic gravitational frequency.

In practical astrophysical units, the velocity expression becomes:

$$v_g^2(r) \text{ km}^2/\text{s}^2 = 80.861 \frac{\rho_c}{10^{-24} \text{ g/cm}^3} \frac{R}{\text{kpc}}^2 \times (1 - \Omega^2) \frac{\sin(\pi r/R) - \cos \frac{\pi r}{R}}{\frac{\pi r}{R}} + \frac{\Omega^2}{3} \frac{\pi r}{R}^2. \quad (11)$$

This model effectively captures the influence of both self-gravity and rotation on the halo structure, allowing for rotation curve fitting using parameters ρ_c , Ω , and R .

2.8 Bayesian Inference

Bayesian inference offers several advantages over traditional fitting methods such as genetic algorithms or least-squares. First, it provides a full posterior distribution for each parameter, enabling uncertainty quantification through credible intervals and parameter correlations. It also allows the natural incorporation of prior knowledge, for example from previous literature or physical constraints. By marginalizing over parameters instead of relying solely on a single best-fit point, Bayesian methods are more robust against overfitting to noise in the data. Finally, Bayesian evidence provides a rigorous framework for model comparison, making it possible to quantitatively assess the relative support for competing models, such as BEC versus NFW.

2.9 Bayesian Model Comparison

In Bayesian statistics, model comparison is based on the *Bayesian evidence* (or marginal likelihood) Z , defined as

$$Z = \int L(\theta) \pi(\theta) d\theta, \quad (12)$$

where $L(\theta)$ is the likelihood of the data given parameters θ , and $\pi(\theta)$ is the prior. Models with larger $\ln Z$ are more strongly supported by the data. The difference $\Delta \ln Z$ between two models is directly related to the Bayes factor, quantifying how much more the data favor one over the other. Following Jeffreys' scale, $\Delta \ln Z \gtrsim 2$ indicates positive evidence, while $\Delta \ln Z \gtrsim 5$ constitutes strong evidence.

Directly evaluating the integral for Z is generally infeasible for high-dimensional parameter spaces, as encountered in halo modeling. We therefore employed two complementary approaches: a simple approximation based on the Bayesian Information Criterion (BIC) and a posterior-based estimator using the harmonic mean [12]. The combination of these methods allows us to balance computational efficiency with a more faithful use of the full posterior.

2.10 BIC Approximation

The Bayesian Information Criterion (BIC) provides a fast and widely used approximation to the evidence, relying only on the maximum likelihood and the number of free parameters:

$$\text{BIC} = k \ln n - 2 \ln \hat{L},$$

where k is the number of free parameters, n the number of datapoints, and $\hat{L}$ the maximum likelihood. The evidence is then approximated as

$$\ln Z_{\text{BIC}} \approx -\frac{1}{2} \text{BIC}.$$

Although computationally efficient, the BIC does not incorporate the full posterior distribution and tends to favor simpler models, making it best interpreted as a coarse proxy for $\ln Z$.

2.11 Harmonic-Mean Estimator

To go beyond the limitations of the BIC, we also compute the evidence using the harmonic-mean estimator [13], which directly exploits the posterior samples produced by Markov Chain Monte Carlo (MCMC):

$$\ln Z_{\text{harm}} \approx -\ln \frac{1}{N} \sum_{i=1}^N \exp -\ln L(\theta_i),$$

where $\{\theta_i\}$ are N posterior draws.

This method has the advantage of using the entire posterior landscape, providing a more faithful reflection of parameter volume and uncertainty than the BIC, which relies only on a single best-fit point. However, the harmonic-mean estimator is also known to be statistically noisy and can become unstable if the posterior has long tails or is insufficiently sampled.

In the context of this work, where we analyze a large number of galaxies, the harmonic-mean estimator offers a practical balance: it can be applied uniformly across the sample using the existing MCMC chains, and it provides a consistent basis for comparing the relative plausibility of BEC, NFW, and gNFW halo models. By reporting both $\ln Z_{\text{harm}}$ and $\ln Z_{\text{BIC}}$, we use the BIC as a simple analytic cross-check while relying primarily on the harmonic-mean estimator for our conclusions. In this way, the more accurate posterior-driven evidence is supported by, and shown to be consistent with, the approximate criterion.

2.12 Model Parameters and Priors

Each halo model introduces a set of free parameters that, together with the baryonic components, determine the predicted rotation curve. Table 1 summarizes the parameters used in the NFW, gNFW, and BEC (non-rotating and rotating) halo models, along with their physical interpretation and prior distributions adopted in the Bayesian inference.

Table 1: Free parameters in the different halo models and their priors.

Model	Parameter	Meaning	Prior
NFW	$\log M_h$	Halo virial mass	U(8, 13)
	$\log c$	Halo concentration	U(0.3, 1.0)
	$\log Y_d$	Disk mass-to-light scaling	N(-0.3, 0.1)
	$\log Y_b$	Bulge mass-to-light scaling	N(-0.15, 0.1)
gNFW	$\log M_h$	Halo virial mass	U(8, 13)
	$\log c$	Halo concentration	U(0.3, 1.0)
	α	Inner slope parameter	U(0.0, 1.5)
	$\log Y_d$	Disk mass-to-light scaling	N(-0.3, 0.1)
	$\log Y_b$	Bulge mass-to-light scaling	N(-0.15, 0.1)
BEC (non-rot.)	$\log \rho_c$	Central density of condensate	U(-26, -22)
	R_{BEC}	Condensate halo radius (kpc)	U(0, 30)
	$\log Y_d$	Disk mass-to-light scaling	N(-0.3, 0.1)
	$\log Y_b$	Bulge mass-to-light scaling	N(-0.15, 0.1)
BEC (rot.)	$\log \rho_c$	Central density of condensate	U(-26, -22)
	R_{BEC}	Condensate halo radius (kpc)	U(0, 30)
	Ω	Dimensionless rotation parameter	U(0, 1)
	$\log Y_d$	Disk mass-to-light scaling	N(-0.3, 0.1)
	$\log Y_b$	Bulge mass-to-light scaling	N(-0.15, 0.1)

The choice of priors is a crucial element of Bayesian inference, as it encodes our prior knowledge and constrains the physically plausible parameter space. For the halo virial mass M_h and concentration c , we adopt uniform priors in logarithmic space. This choice reflects the fact that both parameters vary over several orders of magnitude across galaxies, and a log-uniform prior ensures equal weighting per decade without biasing the inference toward higher or lower values. Similarly, for the gNFW inner slope α we impose a uniform prior between 0 and 1.5, which encompasses the full range from cored ($\alpha \approx 0$) to cuspy ($\alpha > 1$) profiles while excluding extreme slopes not supported by simulations or observations.

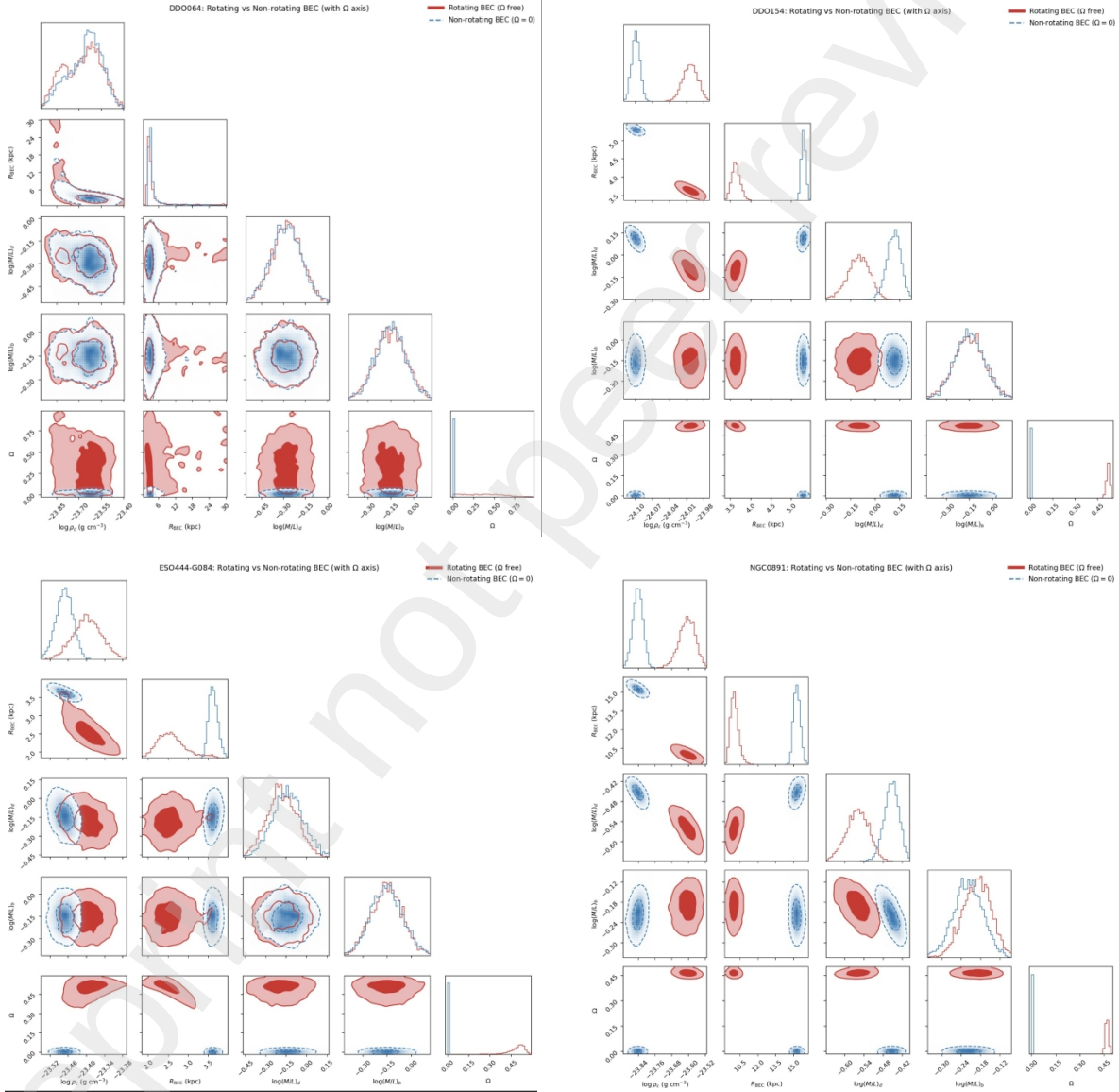
For the BEC models, the central condensate density ρ_c is assigned a log-uniform prior between 10^{-26} and 10^{-22} g cm $^{-3}$, spanning the range of densities considered viable in the literature. The condensate radius R_{BEC} is given a uniform prior up to 30 kpc, which comfortably covers the observed sizes of dwarf and low-surface-brightness

galaxies. In the rotating BEC case, we additionally introduce the dimensionless parameter Ω , restricted to the physically motivated range $[0, 1]$, where $\Omega = 0$ corresponds to a non-rotating condensate and $\Omega = 1$ represents the maximum supportable rotation.

The disk and bulge stellar mass-to-light ratios are modeled with Gaussian priors centered at values motivated by stellar population synthesis studies. Specifically, we adopt $\log Y_d \sim \mathcal{N}(-0.3, 0.1)$ and $\log Y_b \sim \mathcal{N}(-0.15, 0.1)$, which allow for natural variations around the fiducial mass-to-light ratios expected for disks and bulges in the near-infrared.

3 Results

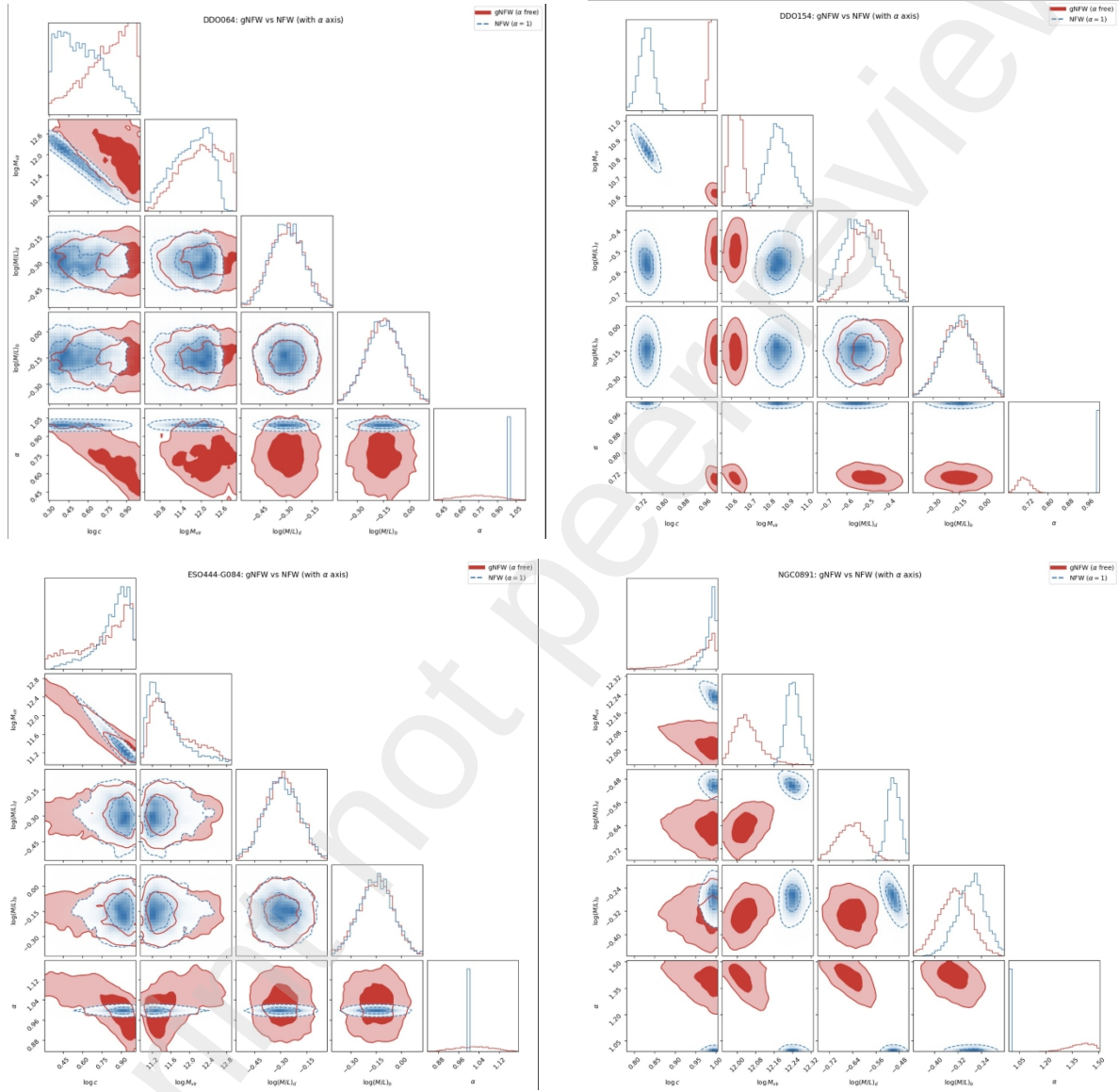
3.1 Posterior Distribution Analysis



(a) BEC model posteriors (64, 154, 444, 891). Red = rotating (Ω free), blue dashed = non-rotating ($\Omega = 0$).

Figure 1: Four-up comparison for the BEC model.

The corner plots show joint posterior distributions comparing the rotating (red, filled) and non-rotating (blue, dashed) BEC halo models, with a one-dimensional marginal for the rotation parameter Ω plotted alongside the joint panels. The main halo parameters are the central condensate density ρ and the condensate radius R , and these are sampled together



(a) NFW (dashed, blue) vs gNFW (filled, red) combined posteriors (64, 154, 444, 891).

Figure 2: Four-up comparison panels per subfigure. Left: BEC rotating vs non-rotating. Right: NFW vs gNFW.

with the disk and bulge mass-to-light scalings Y_d and Y_b under the priors listed in Table I. An elongated degeneracy between ρ and R is evident, reflecting that different combinations of central density and halo radius can produce similar enclosed mass profiles and therefore similar rotation curves. In many panels the red and blue contours largely overlap, indicating that including rotation does not substantially change the inferred halo structure for those galaxies, while in cases where the posterior peaks shift systematically between rotating and non-rotating fits the rotation parameter effectively trades central density for radius to reproduce the kinematics. The Ω marginal is the direct diagnostic of rotation: a posterior peaking clearly away from $\Omega = 0$ (with a narrow credible interval) supports a dynamically rotating condensate, whereas a posterior concentrated near zero implies the non-rotating solution is sufficient.

The second corner plots show joint posterior distributions comparing the NFW (dashed, blue) and gNFW (filled, red) halo models for each galaxy, with the baryonic mass-to-light scalings Y_d and Y_b sampled concurrently with the halo parameters. The gNFW model introduces a free inner slope α (recovering the standard NFW when $\alpha = 1$), while both models include virial mass M_h and concentration c as primary halo parameters per the priors in Table I.

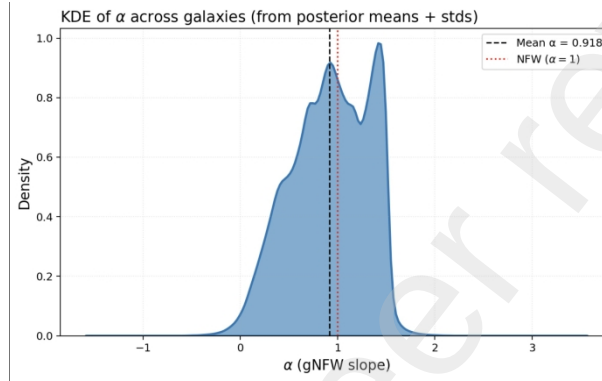


Figure 3: Kernel density estimate (KDE) of the best-fitting inner slope α across our galaxy sample.

The posterior for α is the direct diagnostic of cusp versus core: $\alpha = 1$ corresponds to the canonical NFW cusp, whereas $\alpha < 1$ indicates a shallower, cored inner slope. We also obtain the KDE of best-fitting α across the sample of galaxies (Figure 3). Although the mean value of $\alpha = 0.918$ lies close to the canonical NFW value of 1, the broad range from cored to cuspy profiles still demonstrates significant galaxy-to-galaxy variation which supports the use of a generalized NFW (gNFW) profile with a free α parameter.

3.2 Model Comparison Metrics

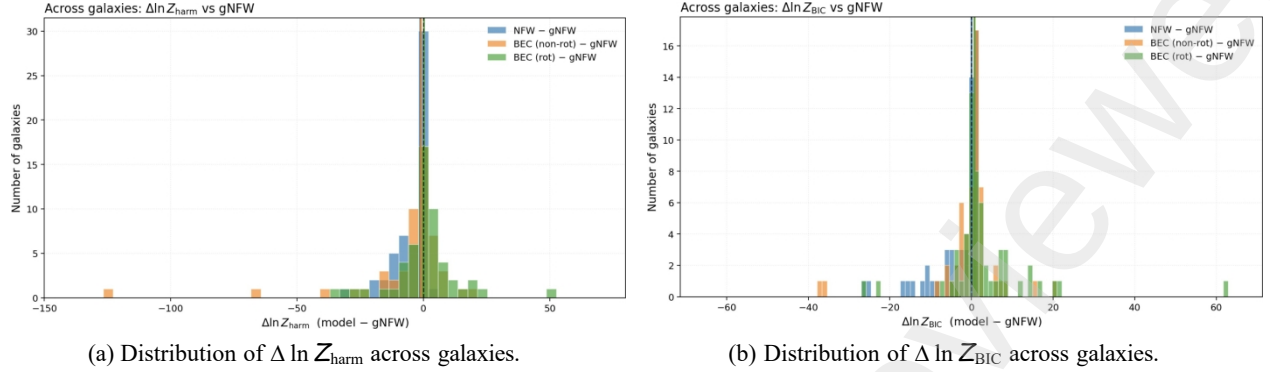


Figure 4: Histograms of evidence differences relative to gNFW. Left: Harmonic mean approximation. Right: Bayesian Information Criterion (BIC). Positive values indicate preference over gNFW.

Figure 4 presents the distributions of log-evidence differences relative to the gNFW model. Panel (a) shows the harmonic-mean approximation, $\Delta \ln Z_{\text{harm}} = \ln Z_{\text{model}} - \ln Z_{\text{gNFW}}$, and Panel (b) shows the Bayesian Information Criterion, $\Delta \ln Z_{\text{BIC}} = \ln Z_{\text{model}} - \ln Z_{\text{gNFW}}$. Blue bars correspond to NFW-gNFW, orange to non-rotating BEC-gNFW, and green to rotating BEC-gNFW. Positive values indicate preference for the alternative model over gNFW.

In both metrics the bulk of galaxies cluster tightly around $\Delta \ln Z = 0$, indicating that for many systems the models are statistically indistinguishable. However, the distributions are asymmetric: for NFW versus gNFW there is a systematic tail toward negative values, showing that allowing the inner slope α to vary yields a modest evidence gain for many galaxies. The BEC models display much broader spreads, with a few galaxies strongly favouring BEC but many exhibiting large negative $\Delta \ln Z$ values, implying a substantial evidence loss compared to gNFW.

Comparing the two panels highlights the effect of the metric itself. The harmonic-mean approximation produces heavier tails, including some extreme outliers, whereas the BIC differences are narrower and more tightly concentrated near zero because the BIC imposes a stronger penalty for additional parameters. Taken together, these results reinforce that a free inner slope provides a small but consistent improvement over a fixed-slope NFW, while BEC models are less robust across the sample and show highly variable performance from galaxy to galaxy even though the rotating BEC has the best mean on both metrics.

Table 2: Evidence-summary statistics relative to gNFW (median [IQR], harmonic-mean evidence).

Model	$\Delta \ln Z_{\text{harm}}$
NFW (fixed slope)	-1.5 [-6.4, 0.7]
BEC (non-rotating)	-1.9 [-8.9, 2.3]
BEC (rotating)	+1.1 [-1.1, 7.8]

The summary statistics in Table 2 show that, relative to gNFW, the fixed-slope NFW profile generally underperforms, with a median $\Delta \ln Z_{\text{harm}} \sim -1.5$ and an interquartile range extending to about -6.4. This indicates that for the majority of galaxies allowing the inner slope to vary gives a small but consistent evidence gain.

In contrast, the non-rotating BEC model exhibits a broader and more asymmetric distribution. Its median evidence difference is similar (about -1.9), but the wide IQR (-8.9 to +2.3) reflects the model's highly variable performance—some galaxies show modest improvements over gNFW while many incur a significant evidence loss.

The rotating BEC model displays a positive median $\Delta \ln Z_{\text{harm}}$ of about +1.1 and a wide upper quartile reaching $\sim +8$, implying that for a subset of galaxies the inclusion of rotation dramatically improves the fit. However, the large spread and occasional extreme outliers underscore that this gain is not universal; in many systems rotation adds complexity without a corresponding evidence increase.

Taken together, these results suggest that flexibility in the inner slope (gNFW) yields the most robust population-level improvement, whereas BEC halos—especially when rotation is included—can outperform gNFW for certain galaxies but at the cost of large variability.

4 Discussion

This analysis compares NFW, gNFW, and BEC (with and without rotation) halo models on SPARC rotation curves within a single Bayesian framework. Two lines of evidence drive our interpretation: (i) the distributions of evidence differences relative to gNFW (Fig. 4) and their summary statistics (Table 2); and (ii) the posterior behaviour of the inner slope parameter in gNFW, summarised by the sample KDE of best-fitting α (Fig. 3), which peaks near but slightly below the canonical NFW value of 1.

The evidence summaries in Table 2 show that, relative to gNFW, the fixed-slope NFW model generally underperforms across the sample, with a median $\Delta \ln Z_{\text{harm}} \approx +1.5$ and an interquartile range extending down to about -6.4 . This pattern is also visible in the left panel of Fig. 4 where the NFW–gNFW histogram skews toward negative values. The result indicates that allowing a free inner slope yields a modest but systematic improvement in fit quality for a substantial fraction of galaxies and this is also shown through the KDE of α .

The BEC models exhibit broader and more heterogeneous behaviour. The non-rotating BEC has a median similar in magnitude to NFW (about -1.9), but its interquartile range is wider $\approx [-8.9, +2.3]$, indicating considerable galaxy-to-galaxy variability. Introducing rotation into the BEC model shifts the median to positive values $\approx +1.1$ and produces a long positive tail (IQR roughly $[-1.1, +7.8]$), showing that a subset of galaxies benefit substantially from rotational support in the condensate model relative to the NFW. At the same time, the distribution remains broad: the positive gains are not universal and several systems still strongly disfavour BEC relative to gNFW.

The inner-slope diagnostics from gNFW are consistent with these evidence patterns. The KDE of best-fitting α across the sample peaks near $\alpha \approx 0.918$, close to the NFW value of 1 but spanning both cored ($\alpha < 1$) and cusp-like ($\alpha \geq 1$) regimes. This dispersion in preferred inner slopes offers a natural explanation for why a flexible profile (gNFW) outperforms a fixed cusp (NFW) in median evidence, and also why BEC—which can mimic a core in some systems—occasionally achieves large positive evidence differences yet remains inconsistent across the full sample.

Methodologically, we report both a posterior-based harmonic-mean evidence and a BIC-derived approximation. The harmonic-mean approach used here leverages the full set of posterior samples and, in our implementation, employs a learned density model to stabilise the estimator; it is therefore more computationally involved but directly tied to the sampled posterior mass. By contrast, the BIC is a simple large-sample approximation based on the maximum likelihood and parameter count. Because the assumptions underlying the BIC-to-evidence mapping need not hold exactly for these data, agreement between the two should be interpreted as a coarse consistency check rather than a requirement. In practice, the qualitative ranking seen with the harmonic-mean estimator is reflected, though more weakly and with narrower tails, in the BIC-derived differences.

Two caveats are especially relevant. First, the treatment of stellar mass-to-light ratios can influence inner halo inferences. Within the scope of this work we adopt informative priors on Y_d and Y_b ; broadening these priors provides a direct robustness test and is a natural extension. Second, other observational systematics (e.g. non-circular motions, bar/warp signatures, beam smearing) are not explicitly modelled here; the SPARC products do not by themselves allow uniform corrections across the full sample. A dedicated follow-up using higher-resolution inner kinematics and explicit inner-radius quality cuts would help quantify their impact. We do not present a formal correlation analysis between α and global galaxy properties here. Methodologically, a Bayesian treatment provides an appropriate route for such investigations.

5 Conclusions

Within a common Bayesian framework applied to SPARC rotation curves, we find that allowing a free inner slope (gNFW) provides a modest but systematic evidence gain over fixed-slope NFW on average, consistent with the sample KDE of α peaking near ~ 0.92 with substantial spread to both cored and cusp-like values. Bose–Einstein condensate halos exhibit heterogeneous performance: the non-rotating model shows wide scatter with a slightly negative median relative to gNFW, while the rotating variant achieves a positive median and a pronounced positive tail, indicating strong improvements for a subset of galaxies but not uniformly across the sample. These results favour flexible inner profiles at the population level and suggest that condensate rotation can play a role in individual cases, though with significant galaxy-to-galaxy variability.

Future work will extend the analysis along two axes. On the statistical side, we will (i) probe robustness to broader stellar mass-to-light priors and (ii) explore alternative evidence estimators (e.g. bridge or thermodynamic integration, nested sampling) alongside the learned harmonic-mean approach. On the astrophysical side, we will (i) perform a hierarchical population analysis of the α distribution and its dependence on galaxy properties and (ii) incorporate higher-resolution inner kinematics to better control non-circular motions and beam-smearing effects.

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